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Studierichting: Informatica
Oefeningenreeks 1C nr. 45.9

Opgave:

Trek de derdemachtswortel in $\mathbb{C}$: $z^3 = 1 - i$

Oplossing:

$$z^3 = 1 - i = \sqrt{2}\,\mathrm{cis}\left(\frac{7\pi}{4}\right)$$

$$\sqrt[3]{r} = \sqrt[3]{\sqrt{2}} = \sqrt[6]{2}$$

$$\theta = \frac{\frac{7\pi}{12} + k2\pi}{3} \text{ met } k \in \{0, 1, 2\}$$

$$\Rightarrow z = \sqrt[6]{2}\,\mathrm{cis}\left(\frac{\frac{7\pi}{12} + k2\pi}{3}\right) \text{ met } k \in \{0, 1, 2\}$$

$$z_0 = \sqrt[6]{2}\,\mathrm{cis}\left(\frac{7\pi}{12}\right)$$

$$z_1 = \sqrt[6]{2}\,\mathrm{cis}\left(\frac{15\pi}{12}\right) = \sqrt[6]{2}\,\mathrm{cis}\left(\frac{5\pi}{4}\right) = \sqrt[6]{2}\,\mathrm{cis}\left(\frac{-3\pi}{4}\right) = -\frac{\sqrt[6]{16}}{2} - \frac{\sqrt[6]{16}}{2}i$$

$$z_2 = \sqrt[6]{2}\,\mathrm{cis}\left(\frac{23\pi}{12}\right)$$

References